

The OptiSparseMET Breeder's Guide

Designing Sparse Multi-Environment Trials — A Plain-Language
Companion

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1st Edition

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About this guide

OptiSparseMET gives a breeding program ten tools for turning three things — a list of candidate lines, a list of candidate locations, and a fixed budget of plots and seed — into a multi-environment trial you can actually plant. It covers the whole chain: ***which sites to use, which lines to test, which line goes to which site, how many plots each entry gets, where each plot sits in the field, and whether the resulting design is precise enough to be worth planting.***

This guide is written for breeders, trial managers, program leaders, and reviewers who will use or approve a trial design, not only for the statisticians who build them. It is deliberately non-technical and code-free: *every tool is described in terms of the decision it makes, the question it answers, when it helps, and when it does not.* Where a tool rests on an assumption that could quietly mislead you, the guide says so plainly rather than leaving it in the formulae.

1. Part 1 — The big picture

1.1. The decision chain this toolkit sits inside

Every tool in this guide sits downstream of one uncomfortable starting point: you have more candidate lines than you can plant in every location, and fewer plots and less seed than your ambitions require. Sparse testing is the response to that — you deliberately test each line in only some of the locations, and let the relationships among lines and among environments carry the rest of the information.

Once you accept that, a chain of decisions follows, and each one constrains the next:

1. Which locations should the trial use? Not every site you could plant is worth planting, and sites differ in how well they represent the conditions your variety will eventually face.
2. Which lines should be tested at all? If your candidate list is larger than your capacity, the subset you choose determines how well you can predict the rest.
3. Which line goes to which location? This is the sparse allocation itself, and it decides how well any two lines can be compared across sites.
4. How many plots should each entry get, and can your seed stock actually supply them?
5. Where should each plot physically sit within each field, given that soil is never uniform?
6. And finally: is the design that comes out of all this actually better than the simpler one you would otherwise have planted?

The ten tools in this guide answer pieces of that chain. None of them replaces your agronomy, your germplasm knowledge, or your genomic prediction model; they decide how to spend a fixed budget of plots and seed so that the data you get back supports the selection decisions you need to make.

1.2. A map of the ten tools

Stage of the decision	Tool	What it decides
Before anything else	Feasibility & capacity checks	Whether the design you have in mind can exist at all
Which locations	Environment selection	Which sites best represent your target environments
Which lines	Entry selection	Which candidates best represent your germplasm
Who goes where	Sparse allocation	Which line is tested in which environment
Who goes where, refined	Connectivity-optimised allocation	The same, tuned for prediction accuracy
How many plots each	Seed-aware replication planning	Replication levels your seed stock can actually support
Where each plot sits	Within-environment field layout	The plantable field book for each site
A better arrangement	Field-layout optimisation	A searched layout with lower error variance
Is it any good?	Design evaluation & sensitivity	Precision and reliability before you plant
Is it worth it?	Simulation	Whether the design raises accuracy and genetic gain

Part 2 walks through each of these in turn. Part 3 gives a consolidated decision guide, a note on how the tools chain together into a single pipeline, and a glossary of the recurring terms.

1.3. Two levels, not one straight line

It is tempting to read the table above as a single conveyor belt: choose sites, choose lines, allocate, lay out fields, plant. In practice the design problem has two levels that are easy to treat separately and shouldn't be:

- The allocation level — which lines appear in which environments. This governs how well lines can be compared across sites, and therefore how well genotype-by-environment effects can be estimated at all.
- The field level — where each plot physically sits within each environment. This governs how much soil variation contaminates each individual plot measurement.

These two levels multiply rather than add. A beautifully balanced allocation planted in a badly laid-out field yields noisy plot values, and noisy plot values degrade exactly the cross-environment comparisons the allocation was designed to protect. Conversely, an immaculate field layout cannot rescue an allocation in which two sites share almost no lines, because there is simply no information linking them.

OptiSparseMET takes this seriously enough to compute it. Its evaluation tool (Section 2.9) builds a single measure of precision across the whole trial that combines the allocation pattern with how precisely each

individual environment measures a plot. That means you can ask whether a change at either level actually improves the trial as a whole, rather than improving one level and quietly paying for it at the other.

The practical takeaway: decide the allocation first, because it constrains what field layouts are even possible, but evaluate the pair together before you commit.

1.4. A few ideas worth understanding before the rest of the guide

A handful of concepts recur across almost every tool. Getting comfortable with them here makes the rest of the guide much faster to read.

Sparse testing. Rather than planting every line in every location, each line is planted in only some of them. You lose a little precision per line and gain a great deal of coverage — more lines, more sites, or both, for the same plot budget. The trade is usually favourable when your target environments are genuinely diverse, and much less favourable when they behave almost identically, in which case a smaller balanced trial may serve you better.

The target population of environments (TPE) and of genotypes (TPG). The TPE is the full range of conditions your future variety must perform in; the locations you actually plant are a sample of it. The TPG is the full germplasm pool; the lines you actually test are a sample of that. Almost every question in this guide is really a sampling question: does this sample represent that population well enough to support the decision you are about to make?

Connectivity. Two environments are connected when they share tested lines, and two lines are connected when they are tested together somewhere. Connectivity is what makes cross-environment comparison possible at all. It is easy to build an allocation that looks tidy — every line tested the same number of times, every site the same size — while some pairs of sites share far more lines than others. That unevenness weakens exactly the comparisons you care about, and it is invisible unless you look for it directly.

“Equireplicate”, not “balanced incomplete block design”. This deserves a plain statement, because the literature is loose about it. A true balanced incomplete block design (BIBD) requires that every pair of lines occurs together in the same environment exactly the same number of times. That is a very strong property, and for the situation every breeder is actually in — hundreds or thousands of lines across a handful of environments — it is mathematically impossible, not merely difficult. What is achievable is equal replication (every line tested the same number of times), equal environment size, and pairwise co-occurrence made as even as it can be. That is what this package builds, and that is what it now calls the method: equireplicate. If a tool or a paper tells you it has given you a BIBD across environments in that setting, it has not.

Reliability, PEV and CDmean. Prediction error variance (PEV) measures how uncertain your estimate of a line's genetic value is — lower is better. Reliability, often reported as CDmean, rescales that onto a 0-to-

1 scale where higher is better, and is easier to reason about: roughly, the proportion of the genetic signal your trial actually recovers. Both are computed from the design alone, before a single plot is harvested, which is what makes them useful for choosing between designs.

A- and D-optimality. Two classical yardsticks for a field layout. A-optimality asks for the smallest average uncertainty when comparing pairs of entries; D-optimality asks for the smallest overall uncertainty taken as a whole. For both, a lower criterion value is better. They usually agree; when they disagree, A-optimality is the one that matches “I want to compare entries against each other”, which is normally what a breeder wants.

Spatial variation. Soil fertility, drainage and shading vary smoothly across a field, so neighbouring plots are more alike than distant ones. Good layouts spread related material and repeated checks across that gradient rather than letting a family sit together in one corner. How much this matters depends entirely on how variable your fields are — which is why the package lets you state your own level rather than assuming one.

Seed, the constraint most theory ignores. Design textbooks assume you can plant whatever replication the mathematics prefers. Real programs run out of seed for specific lines, and a design that cannot be planted is worth nothing. This package treats seed availability as a first-class input rather than an afterthought — one of the more practically useful things about it.

2. Part 2 — The ten tools

2.1. Feasibility and capacity checks — can this design exist at all?

What it is. A set of fast arithmetic checks that run before any allocation is attempted, telling you whether the trial dimensions you have in mind are internally consistent — and, if you were hoping for exact balance, whether that is even possible.

What question it answers. “Given this many lines, this many sites, and this many plots per site, can the design I am imagining actually be built?”

Breeding-program analogy. This is the seed-store check you do before writing the planting plan, not after: counting whether you have enough bags before promising the field crew a layout.

When to reach for it.

- Always, before the first allocation of a new season — trial dimensions change every year and the arithmetic changes with them.
- Whenever you want equal replication for every line, since that requires the number of available slots to divide exactly by the number of lines.

- When someone asks you for a “balanced” design and you need a defensible answer about whether that is achievable at your scale.

When to skip it.

- Rarely worth skipping — it costs nothing and prevents a class of confusing downstream errors.

How it works in practice. You supply the number of lines, sites, entries per site and desired replication. The check reports whether the slot arithmetic works exactly, and separately reports the stricter conditions a genuine BIBD would require — including whether the number of environments is large enough relative to the number of lines. In realistic breeding settings that last condition fails, and the check tells you so directly instead of letting a tool claim balance it cannot deliver. If a true BIBD genuinely is possible — which happens only in small, special cases — a dedicated constructor can build one.

2.2. Environment selection — which locations represent your TPE

What it is. A method for choosing a subset of locations from a longer list of candidate sites, so that the ones you plant represent the range of conditions your variety must eventually handle.

What question it answers. “I can afford eight sites out of thirty. Which eight?”

Breeding-program analogy. Choosing which regions to survey before a product launch: you want the sample to reflect the market, not just the places easiest to reach.

Coverage, not just spread — a correction worth knowing. There are two different things people mean by a “representative” set of sites, and conflating them causes real mistakes. One is diversity: pick sites as unlike each other as possible. The other is coverage: pick sites such that every site you did not plant is well represented by one you did. These are not the same. A maximally diverse set can consist entirely of extreme, atypical locations — mutually different, and jointly unrepresentative of the ordinary conditions most of your growers face. This package therefore offers coverage as the default criterion, and keeps the diversity-style criterion available but clearly labelled, alongside the clustering and random baselines you would want for comparison.

When to reach for it.

- Your candidate site list is longer than your budget and the sites genuinely differ in climate, soil or management.
- You suspect your historical site list has drifted — several sites now measuring much the same thing, while a distinct production zone goes unsampled.
- You need to justify a site list to a funder or a review panel with something more principled than “these are the stations we have”.

When to be cautious or skip it.

- If your sites are already few and each is there for a non-negotiable reason — a mandated region, a disease nursery — there is no real choice to optimise.
- The method needs some measure of how similar sites are to one another. If that comes from very thin historical data, treat the output as a discussion aid rather than an answer.
- Representativeness is not the same as trial quality. A site that represents the TPE well but is run badly still yields poor data; use this alongside, not instead of, your knowledge of which stations are reliable.

How it works in practice. You supply a matrix describing how similar the candidate sites are — typically from genetic correlations across historical trials, or from environmental data such as weather and soil — and the number of sites you can afford. The tool returns the chosen subset and a coverage score. Running it across a range of subset sizes shows where adding another site stops buying much, which is often more informative than the choice at any single size.

2.3. Entry selection — which lines represent your TPG

What it is. A method for choosing which subset of your candidate lines to actually test, when you cannot phenotype them all, in a way that maximises how well the untested remainder can be predicted.

What question it answers. “I can phenotype 300 of these 1,200 lines. Which 300 teach me the most about the other 900?”

Breeding-program analogy. Choosing which students to interview when you must forecast the whole cohort's performance: you want a set that spans the group, not the 300 who happen to be most similar to each other.

Why not just take the top ranked lines. A tested set chosen purely by predicted merit tends to be genetically narrow — close relatives carrying overlapping information. It will estimate its own members well and predict the rest of the panel poorly. This tool optimises a reliability criterion instead, so the chosen set spans the relationship structure of the panel.

When to reach for it.

- You are assembling a training or calibration set for genomic prediction and want it to serve the whole panel.
- Your candidate list far exceeds your phenotyping capacity and the panel contains obvious family structure.
- You want a defensible, repeatable rule for who gets tested, rather than a negotiated list.

When to skip it.

- If you intend to test everything, there is no subset to choose.

- If the decision is really “which lines advance”, that is a selection question driven by merit, not a representation question — this tool deliberately does not optimise merit.
- Its answers are only as good as the relationship matrix behind them; with poor marker coverage, treat them cautiously.

How it works in practice. You supply a relationship matrix for the candidates and the size of the tested set. The tool searches by exchanging members in and out, keeping changes that improve overall reliability, and returns the chosen set with its score. Running it at several set sizes shows the point of diminishing returns — usually the most actionable output.

2.4. Sparse allocation across environments — who goes where

What it is. The core of the package: given your lines, your sites and your per-site capacity, it decides which lines are tested at which sites.

What question it answers. “I can plant 200 entries per site across 6 sites, but I have 800 lines. Who goes where?”

Breeding-program analogy. Building a fixture list for a league where not every team can play every other: you want the schedule to be fair, and you want enough shared opponents that the final table means something.

Two strategies, for two different situations. A coverage-first strategy guarantees every line is planted somewhere and tolerates sites of unequal capacity — the pragmatic choice when your stations genuinely differ in size. An equireplicate strategy instead guarantees that every line is tested exactly the same number of times and that every site carries exactly the same number of entries. That equal treatment is a real fairness and precision property: no line is advantaged by being measured more often than its competitors.

The balance option, and why it matters more than it looks. Equal replication alone does not guarantee that the design is well connected. You can satisfy it while some pairs of sites share many lines and others share few. The package therefore offers an additional refinement that rearranges which line sits at which site — without disturbing the equal replication or equal site sizes — to even out either the number of lines each pair of sites shares, or how often pairs of lines are tested together. In practice the site-pair version is the one most worth using: it directly strengthens the cross-site comparisons that genotype-by-environment analysis depends on, and it can often be evened out completely.

When to reach for it.

- Any time your candidate list exceeds what you can plant everywhere — which is to say, most seasons.

- When equal replication matters for fairness or for defensibility, use the equireplicate strategy and add the balance refinement.
- When your sites differ substantially in capacity, use the coverage-first strategy instead; forcing equal sizes onto unequal stations wastes plots.

When to be cautious.

- Equireplicate requires the arithmetic to work out exactly; run the feasibility check first (Section 2.1) rather than discovering this at allocation time.
- This tool balances structure, not prediction. If your real objective is prediction accuracy under genotype-by-environment interaction, the next tool optimises that directly.
- If you are also enforcing family or genetic-group constraints, the balance refinement stands aside so it does not break them; decide which property matters more to you.

How it works in practice. You supply the lines, the sites, the entries per site, and how many sites each line should appear in. You may also nominate common entries — checks or reference lines planted at every site — which provide a simple, model-free backbone of connectivity and are usually worth including. The tool returns the allocation, a summary of replication and site sizes, and the matrix of how often each pair co-occurs, so you can see the connectivity you actually got rather than assuming it.

2.5. Connectivity-optimised allocation — allocating for prediction, not just for balance

What it is. A refinement step that takes an existing allocation and rearranges it to directly maximise how precisely the trial will predict genetic values across your target environments.

What question it answers. “My allocation is structurally tidy. Is it the allocation that actually predicts best?”

Breeding-program analogy. The difference between a fixture list that is fair on paper and one deliberately arranged so the final standings are as informative as possible.

Why this is a different question from Section 2.4. Structural balance is a proxy: we believe even replication and even co-occurrence lead to good prediction. This tool skips the proxy and optimises the outcome itself — the expected precision of across-environment genetic values under a genotype-by-environment model that includes how correlated your sites are. Where those correlations are uneven, the best-predicting allocation is not always the most symmetric one.

When to reach for it.

- Your target environments differ substantially in how they rank genotypes — a broad or heterogeneous TPE, where allocation choices have the most leverage.
- Genomic prediction across sites is a primary output of the trial, not an afterthought.

- You have a usable estimate of how your environments correlate, from historical multi-environment data or environmental covariates.

When to skip it.

- If your sites are highly correlated — a narrow TPE — the gains are small and a balanced allocation is essentially as good.
- It needs an environment correlation structure. Without a credible one, you are optimising against a guess, and structural balance is the safer default.
- It evaluates a full precision calculation at every step, so very large trials need patience or a coarser starting point.

How it works in practice. You hand it an allocation (typically from Section 2.4), a relationship matrix for the lines, and an environment correlation matrix. It exchanges entries between sites — never disturbing replication counts or site sizes — and keeps changes that improve predicted precision, reporting the before-and-after figures so you can see whether the effort bought anything.

2.6. Seed-aware replication planning — how many plots each entry really gets

What it is. A planner that converts your intended replication into one your seed store can actually supply, and, separately, a way to work out what replication level is worth aiming for in the first place.

What question it answers. “How many plots should each entry get — and can I actually plant that?”

Breeding-program analogy. The moment the elegant plan meets the seed store, and some lines turn out to have one packet left.

Two connected jobs. The first is feasibility: given seed on hand and seed required per plot, decide which entries can carry full replication, which drop to a single plot, and which cannot be planted at all — so the design that leaves the office is one the field crew can execute. The second is strategy: sweep across candidate replication levels and report the accuracy and genetic gain each would deliver at your own level of field variability, so the choice is evidence-based rather than habitual.

A finding worth knowing, and its caveat. With genomic relationships in the analysis, replication often matters less than breeders expect: testing more unique entries once frequently beats testing fewer entries twice, because relatives contribute information to each other. But this depends on how variable your fields are and how strong your relationship information is. Treat it as a hypothesis to check for your program, not a rule — which is exactly what the sweep is for.

When to reach for it.

- Any season where seed is genuinely limiting for some entries — early-generation material, new introductions, a difficult increase.

- When deciding between “more entries, less replication” and the reverse, and you want the trade-off quantified for your own conditions.
- When a design must be defensible as plantable, not merely statistically attractive.

When to skip it.

- If seed is abundant and replication is fixed by protocol, the feasibility step has nothing to do.
- The strategy sweep assumes you can state your field variability roughly; if you cannot, run it across a range and see whether the recommendation is stable.

How it works in practice. For feasibility, supply seed available per line and seed needed per plot; the planner partitions entries into fully replicated, single-plot and excluded, using a rule you choose. For strategy, supply your relationship matrix and variance assumptions and a set of candidate replication levels; it reports accuracy, genetic gain and precision at each, flags any level your seed cannot support, and recommends where returns flatten out.

2.7. Within-environment field layout — turning an entry list into a plantable field

What it is. The step that converts “these entries are grown at this site” into an actual field book: rows, columns, blocks, and which entry sits in which plot.

What question it answers. “Where does each plot physically go?”

Breeding-program analogy. The seating plan, not the guest list. Who is invited was decided earlier; this decides who sits next to whom, and it matters because the room is not uniform.

Two engines, for two situations. A block-based engine handles designs built around repeated checks — augmented layouts where most entries appear once and checks recur in every block, partially replicated layouts where a fraction of entries repeat, and conventional replicated block structures. A row-and-column engine handles fixed grids where controlling variation along both field axes simultaneously is the priority. Which you want is usually dictated by how much seed you have and how variable the site is, rather than by preference.

Spreading genetic material deliberately. Both engines can take family labels or a relationship matrix and use them to avoid placing close relatives together. This matters because the most common field-layout mistake is planting a family as a contiguous block for convenience — which confounds that family's genetics with whatever soil gradient runs beneath it, in a way no analysis can fully undo.

When to reach for it.

- Every environment in every trial — some layout has to be chosen, and choosing it deliberately costs nothing.

- Especially when your fields have known gradients, or when entries arrive grouped by family and would otherwise be planted in that order.

When to be cautious.

- The layout must accommodate exactly the entries the allocation assigned; if your field dimensions do not match the plot count, the tool will adjust them and tell you it did. Read that message rather than ignoring it.
- Checks consume plots. Enough to anchor the spatial model, not so many that they crowd out the entries you are trying to evaluate.

How it works in practice. You supply the entries for that site, your check lines, the field dimensions and the block structure you want, optionally with family or relationship information. The tool returns a field book with each plot's row, column, block and entry, ready to hand to the field crew or to combine with the other sites into a single trial-wide book.

2.8. Field-layout optimisation — searching for a better arrangement

What it is. A search that starts from a valid layout and rearranges entries within the field, keeping arrangements that score better on a precision criterion you choose.

What question it answers. “Of the many valid layouts for this field, which one measures my entries most precisely?”

Breeding-program analogy. Reshuffling the seating plan a few hundred times and keeping the arrangement where the most useful conversations can actually happen.

Choosing the criterion. Optimise A-optimality when your priority is comparing entries against one another — usually the right default for variety evaluation. Optimise reliability when genomic prediction of untested or partially tested material is the goal. Optimise D-optimality when overall estimation quality matters more than specific pairwise comparisons. The three usually broadly agree; where they diverge, the choice should follow what the trial is for.

When to reach for it.

- Sites with strong or well-documented spatial variation, where layout has the most to gain.
- High-stakes trials — final-stage evaluation, registration data — where the extra computing time is trivial against the cost of a poor decision.
- When you must document that the layout was chosen on a stated criterion rather than by convenience.

When to skip it.

- Uniform sites with low spatial variation, where a sensible randomised layout is already close to optimal and the gain does not repay the effort.
- Under time pressure: a good constructed layout from Section 2.7 is far better than no design, and the marginal gain here is usually modest.
- Very large fields, where the search becomes slow — narrow the problem or accept the constructed layout.

How it works in practice. You supply a layout, your criterion, your variance assumptions, and how long to search. Several search strategies are available, trading thoroughness against time. The tool reports the best layout found and the score trace, so you can see whether the search had converged or was still improving when it stopped — worth checking before trusting the result.

2.9. Design evaluation and sensitivity — checking a design before you plant it

What it is. The measurement instrument for the whole package: it takes a design and reports how precisely it will estimate what you care about — before a single plot is sown.

What question it answers. “How good is this design, in numbers, and how much do those numbers depend on assumptions I am not sure about?”

Breeding-program analogy. A structural survey before construction. It will not tell you whether you should have wanted the building; it tells you whether this one will stand up.

Two scales of evaluation. Within a single environment, the tools report A-optimality, D-optimality and reliability for a given field layout under your assumed spatial structure. Across the whole trial, a second tool combines the allocation pattern with how precisely each site measures a plot, and reports the reliability of genetic values across your target environments. That across-trial figure is the one to quote when comparing designs, because it is the only one that reflects both levels at once.

Sensitivity — the part most often skipped. Every one of those numbers is conditional on assumed variance components, and programs rarely know theirs precisely. A sensitivity sweep re-computes precision across a range of plausible values, so you can see whether your preferred design is genuinely better or only better under one particular guess. A design that wins across the whole range is a safe choice; one that wins only at a specific assumption is a warning.

When to reach for it.

- Whenever you are choosing between two or more candidate designs — this is what converts an argument into a comparison.
- Before committing a large or high-stakes trial, as due diligence.
- Whenever your variance assumptions are uncertain, which is most of the time.

When to be cautious.

- These are model-based predictions of precision, not guarantees about your season. They tell you which design is better positioned, not what your heritability will be.
- Compare designs on the same assumptions. Comparing two designs evaluated under different variance settings tells you nothing.

How it works in practice. You supply the design and your variance assumptions, and receive the criteria described above. For the across-trial figure you also supply a relationship matrix and an environment correlation structure. Because the calculation is fast, the honest workflow is to evaluate several candidate designs and keep the comparison, rather than evaluating one and hoping.

2.10. Simulation — proving the design earns its keep

What it is. A simulation that generates data under a genotype-by-environment model, analyses it as you would analyse the real trial, and reports the two outcomes breeders actually care about: selection accuracy and genetic gain.

What question it answers. “Does this design actually select better lines than the one I would otherwise have planted?”

Breeding-program analogy. A dress rehearsal with a known answer. Because the simulation knows each line's true genetic value, it can tell you exactly how often your design would have picked the right ones.

Why this exists, and why it matters. Every other evaluation in this guide reports a design criterion — a proxy. Proxies are chosen because they are believed to track selection quality, but that belief is worth testing rather than assuming. This tool closes the loop: it takes the design you are about to plant and reports how accurately it would rank genotypes and how much genetic gain a given selection intensity would deliver. If a design improves the criteria but not accuracy, you have learnt something important before spending a season on it.

When to reach for it.

- When deciding whether sparse testing is worth adopting at all, compared with a smaller balanced trial — the single most valuable use of this tool.
- When a design change is expensive or politically contested and you want evidence rather than assertion.
- When setting program-level policy — how many sites, how sparse, how much replication — that will apply for several seasons.

When to be cautious.

- Simulation results reflect the genetic architecture you simulate. If your assumed environment correlations are wrong, the conclusions inherit that error — so vary them and check whether your conclusion survives.
- It answers relative questions well (“is A better than B?”) and absolute ones poorly (“what accuracy will I get?”). Use it to choose, not to forecast.
- Several replicates are needed for a stable comparison; a single run tells you little.

How it works in practice. You supply the design, a relationship matrix, an environment correlation structure and your variance assumptions, and choose how many replicate simulations to run. The tool reports mean accuracy and genetic gain with their variability. The most informative use is comparative: run the same simulation on a balanced design, a random sparse design and your optimised design, and read the differences.

3. Part 3 — Putting it together

3.1. The end-to-end pipeline

The tools in Part 2 are described separately because each answers a distinct question, but in a normal season they run in sequence, and the package provides a single pipeline that chains them: allocate entries across environments, apply the seed-aware replication plan, build the field layout for every site, and assemble one combined field book covering the whole trial.

For routine seasons the pipeline is the sensible entry point — it enforces the ordering and passes each stage's output into the next without transcription errors. Reach for the individual tools when you need to interrogate a specific decision: comparing site subsets, testing whether the allocation is well connected, or evaluating two candidate layouts against each other. A reasonable habit is to run the pipeline for the design you intend to plant, then use the evaluation and simulation tools to check that the result deserves to be planted.

3.2. Decision guide

If your question is...	Reach for...
"Can I even build the design I sketched?"	Feasibility & capacity checks
"I have 30 possible sites and budget for 8."	Environment selection
"Which lines should form my tested set?"	Entry selection
"I can't plant every line everywhere."	Sparse allocation
"My allocation is balanced, but is it well connected?"	Sparse allocation's balance option
"I want the allocation that predicts best, not just the tidiest."	Connectivity-optimised allocation
"I don't have enough seed for two reps of everything."	Seed-aware replication planning
"Turn this entry list into a field I can plant."	Within-environment field layout
"My field has a strong fertility gradient."	Field-layout optimisation
"How precise will this trial actually be?"	Design evaluation & sensitivity
"Is sparse testing actually worth it for my program?"	Simulation

3.3. A rough guide by program shape

Programs testing large candidate numbers across several diverse locations — typical of mid- to late-stage variety evaluation — get the most from this package as a whole. Sparse allocation with the connectivity refinement is the centre of gravity, environment selection matters because sites genuinely differ, and simulation is worth running once to establish how sparse the program can safely go.

Programs with a few well-established sites and no real choice about locations can skip environment selection entirely and focus on allocation, seed-aware replication and field layout. The gains here come from connectivity and from not planting families in contiguous blocks, rather than from site choice.

Early-generation programs with severe seed limitation should start at seed-aware replication planning and work outward. When most entries have seed for a single plot, the design question is really "how many unique entries can I evaluate and how do I lay them out", which points to augmented or partially replicated layouts with careful spatial control.

Programs building or refreshing a training population for genomic prediction should look first at entry selection, then at allocation optimised for prediction. Here representation of the germplasm pool matters more than balance for its own sake.

Programs under regulatory or registration scrutiny should lean on the evaluation and sensitivity tools regardless of shape. Being able to state the criterion a design was chosen on, and show it holds across a range of plausible assumptions, is often as valuable as the design itself.

3.4. A word on running more than one tool

None of these tools is a strict improvement on the others, and several answer genuinely different questions. Where two could apply, running both and comparing is usually more informative than

agonising over which is correct — the comparison itself tells you how much the choice matters for your data. If a balanced allocation and a connectivity-optimised one give nearly identical predicted reliability, that is a useful finding: it means you can plant the simpler one and spend your attention elsewhere.

One habit is worth adopting above all others. Whatever design you arrive at, evaluate it against the simplest alternative you would otherwise have planted, on the same assumptions. Sophistication is not self-justifying, and a tool that cannot demonstrate an advantage over the obvious baseline has not earned its place in your season.

Glossary

A-optimality. A design criterion measuring the average uncertainty when comparing pairs of entries. Lower is better. The natural choice when the trial exists to compare candidates against each other.

Allocation. The decision of which lines are tested in which environments — the incidence pattern of the sparse trial.

Augmented design. A layout in which most entries appear once and a small set of checks is repeated throughout, allowing spatial correction when seed is too limited to replicate entries.

BIBD (balanced incomplete block design). A design in which every pair of treatments occurs together the same number of times. Generally impossible across environments when lines greatly outnumber sites; see equireplicate.

CDmean / reliability. A 0-to-1 measure of how much of the genetic signal a design recovers. Higher is better; easier to interpret than prediction error variance.

Common treatments / checks. Entries planted at every site, providing a simple backbone of connectivity between environments that does not depend on a model.

Concurrence. How often two lines are tested together in the same environment, or how many lines two environments share. The raw material of connectivity.

Connectivity. The extent to which environments and lines are linked through shared testing, determining whether cross-environment comparisons are supportable.

D-optimality. A design criterion measuring overall estimation uncertainty taken as a whole rather than pairwise. Lower is better.

Equireplicate. A design in which every line is tested the same number of times and every environment holds the same number of entries. Achievable in practice, unlike a full BIBD, and what this package builds.

G×E (genotype-by-environment interaction). The phenomenon of genotypes ranking differently across environments — the reason multi-environment testing exists at all.

Genetic gain. The improvement in true genetic merit achieved by selecting on trial results at a given intensity. The ultimate outcome measure.

p-rep (partially replicated) design. A layout in which a fraction of entries are replicated and the remainder appear once — a middle ground between augmented and fully replicated designs.

PEV (prediction error variance). The uncertainty attached to an estimated genetic value. Lower is better; reliability is its rescaled, more interpretable form.

Relationship matrix (GRM). A matrix recording how closely related every pair of lines is, usually built from markers. It allows tested lines to inform predictions for untested relatives.

Selection accuracy. The correlation between predicted and true genetic values — how reliably the trial identifies genuinely superior lines.

Sparse testing. Testing each line in only some environments rather than all, trading a little precision per line for substantially broader coverage.

Spatial variation. Smooth trends in soil or microclimate across a field that make neighbouring plots more alike than distant ones.

TPE (target population of environments). The full range of conditions the eventual variety must perform in; the sites you plant are a sample of it.

TPG (target population of genotypes). The full germplasm pool from which candidates are drawn; the lines you test are a sample of it.

Where to go next

This guide is deliberately non-technical and code-free. For the statistical detail behind every tool described here — the exact criteria, the models they assume, and the validation each has been checked against — see the package documentation and the accompanying technical vignette, which work through a complete example from candidate list to combined field book.

Two habits are worth carrying away. First, state the criterion a design was chosen on, and check it holds across a plausible range of assumptions rather than a single guess. Second, always compare against the simplest design you would otherwise have planted. Both take minutes, and both are the difference between a design you can defend and one you merely prefer.